
MODEL SELECTION AND VALIDATION REPORT (2004_2011_biomass model)

OBSERVATIONS:1643

SANITY: OK - all models in the selection process passed sanity checks.

SELECTED RANDOM TERM: spatiotemporal_random_fields_ar1

MODEL SUMMARY (2004_2011_biomass model)

Spatiotemporal model fit by REML ['sdmTMB']

Formula: fixed_formula

Time column: year

Family: lognormal(link = 'log')

Conditional model:

	coef.est	coef.se
(Intercept)	6.81	0.26
kde_fric	0.22	0.03
kde_feve	-0.26	0.03
tric	0.08	0.02
teve	-0.43	0.03
sbt_mean	0.23	0.07
sbt_sd	0.19	0.05
sst_sd	0.04	0.04
sic_mean	0.11	0.08
log_chla_mean	-0.04	0.03
depth	0.15	0.03

Dispersion parameter: 0.74

Spatiotemporal AR1 correlation (rho): 0.89

Matérn range: 371.38

Spatiotemporal marginal AR1 SD: 0.78

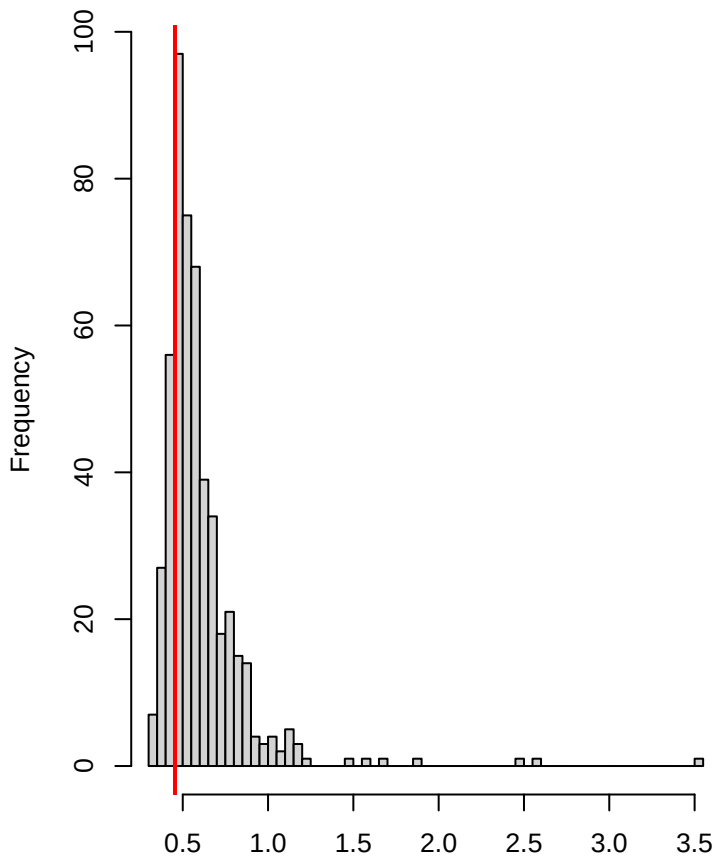
REML criterion at convergence: 12834.181

See ?tidy.sdmTMB to extract these values as a data frame.

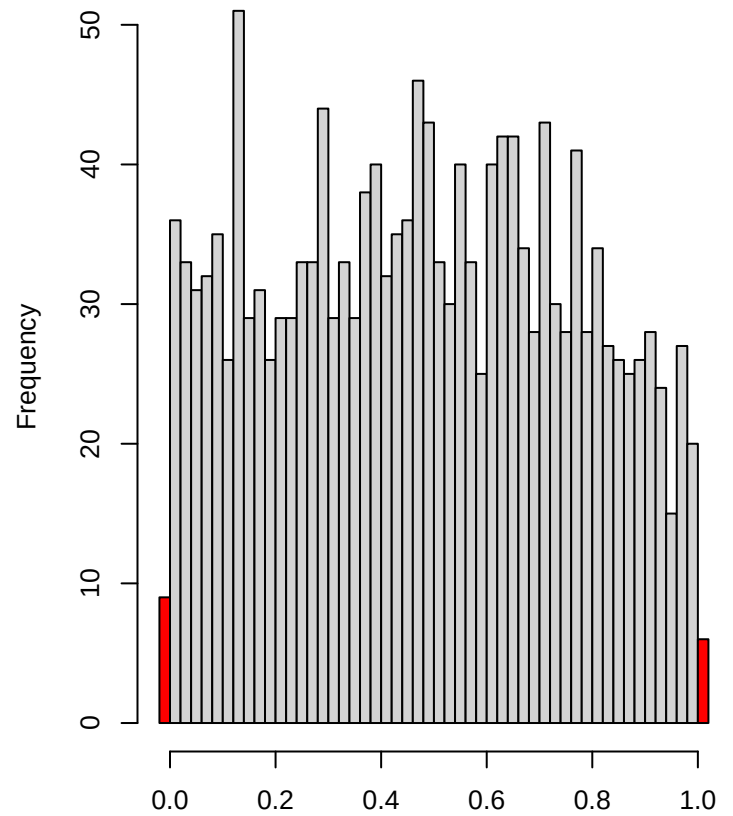
SELECTION STATISTICS

	df	AIC	delta_AIC
spatiotemporal_random_fields_ar1	15	25698.36	1.6679926
spatiotemporal_random_fields_ar1_with_anisotropy	17	25700.03	123.7047546
spatiotemporal_random_fields_iid	14	25823.73	0.6944568
spatiotemporal_random_fields_iid_with_anisotropy	16	25824.43	45.5954050
spatial_random_fields_with_anisotropy	16	25870.02	1.7590544
spatial_random_fields	14	25871.78	486.2559981
error_only	12	26358.04	NA

DHARMA nonparametric dispersion test via sd of residuals fitted vs. simulated

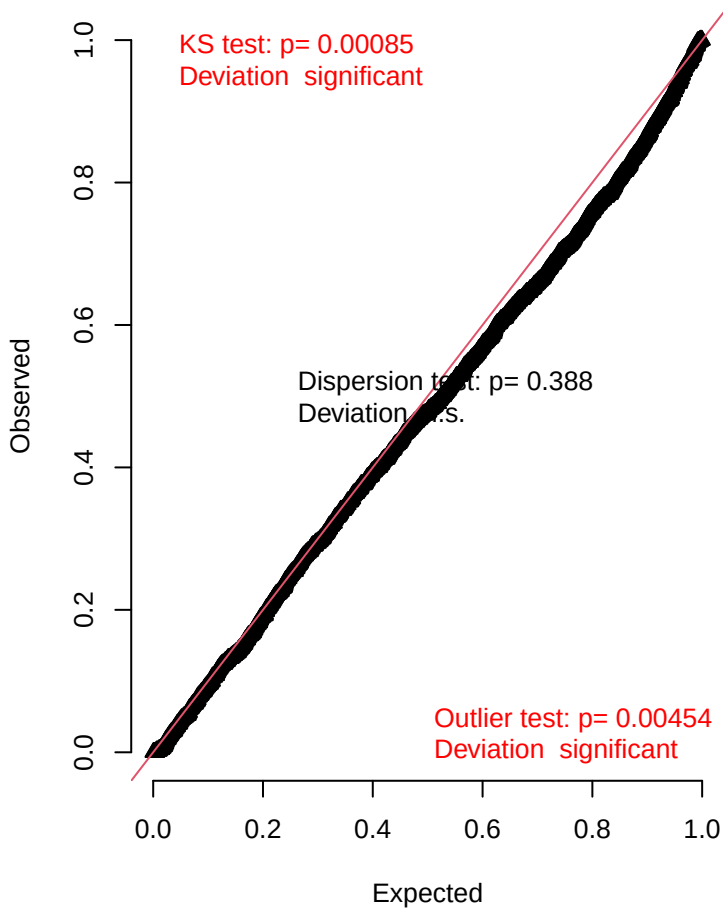


Outlier test significant



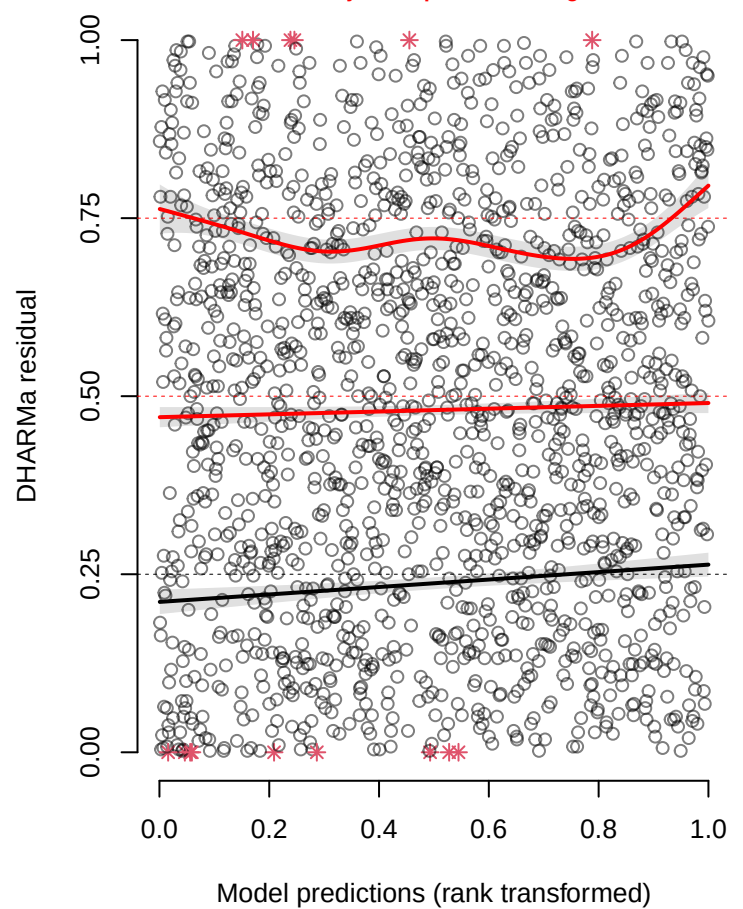
Simulated values, red line = fitted model. p-value (two.sided) =

QQ plot residuals



Residuals (outliers are marked red)

DHARMA residual vs. predicted
Quantile deviations detected (red curves)
Combined adjusted quantile test significant



DHARMa Moran's I test for distance-based autocorrelation

Moran's I = 0.006 | H1: Distance-based autocorrelation | p-value = 0.224

